

Filtering_datasets_final.r

Usuario

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#Title: Filtering_datasets
#Created 29/5/2025
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#Project: NEFINEO_MS.Rproj
#Description: The aim of this code is to get the final globalfungi dataset for its1 and its2.
#1. Create a column to keep samples according to paper id (no plantations).
#2. Create a column to group samples by a threshold of 90m of proximity.
#3. Create a new permanent ID to each sample according to the number of group and ...
#the year of paper to avoid getting grouped samples of different samplings.
#4. Export completed datasets as .csv files
#5. Export short datasets indicating new ID and permanent ID of each sample
#6. Create new ID for new datasets (no global fungi) and write individual .csv for each study

#Set working directory from NEFINEO_MS.Rproj####
#github directory
list.files() #check files in folder

## [1] "Aggregation_PredictorVariables.R" "Buscando_Replicas.R"
## [3] "Data" "Filtering_datasets.pdf"
## [5] "Filtering_datasets.r" "Filtering_datasets_2.R"
## [7] "Filtering_datasets_final.r" "Filtering_datasets_final.spin.R"
## [9] "Filtering_datasets_final.spin.Rmd" "Filtering_datasets_Markdown_EN.Rmd"
## [11] "NEFINEO_MS.Rproj" "README.md"
## [13] "soilgrid_WP2.R" "soilgrid_WP2_newdata.R"

#Datasets in folder: Data

#Open datasets ####
#final dataset its1
its1= read.table("Data/globfungi_metadata_filtered_complemented_its1_soil.tsv",
                sep = "\t",)
length(unique(its1$PermanentID)) #check ID by sample #2815 samples

## [1] 2815

#final dataset of articles to keep in its1 (no plantations)
its1.plant= read.csv("Data/article_list_noplantation.its1.csv")
#final dataset its2
its2= read.table("Data/globfungi_metadata_filtered_complemented_its2_soil.tsv",
                sep = "\t",)
length(unique(its2$PermanentID)) #check ID by sample #2380 samples
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## [1] 2380
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#final dataset of articles to keep in its2 (no plantations)
its2.plant= read.csv("Data/article_list_noplantation.its2.csv")

#Create new vector of articles to keep in datasets its1/its2 ####
#its1
keep.its1= subset(its1.plant, To_keep=="yes", select= title) #articles to keep
nrow(keep.its1) #20 articles to keep
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## [1] 20
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v.keep.its1= match(its1$paper_id, keep.its1$title) #vector with articles to keep
v.keep.its1= ifelse(is.na(v.keep.its1), "no", "yes") #reemplace Na by "no"
#its2
keep.its2= subset(its2.plant, To_keep=="yes", select= title) #articles to keep
nrow(keep.its2) #28 articles to keep
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## [1] 28
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v.keep.its2= match(its2$paper_id, keep.its2$title) #vector with articles to keep
v.keep.its2= ifelse(is.na(v.keep.its2), "no", "yes") #reemplace Na by "no"

#Add new vector of articles to keep to dataset ####
new.its1= cbind(its1[,1:5], paper_to_keep=v.keep.its1, its1[,6:171])
new.its2= cbind(its2[,1:5], paper_to_keep=v.keep.its2, its2[,6:171])

#Create new vector of grouped samples by a threshold of 90m of proximity ####
#install packages if need it
library(dplyr)
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## Attaching package: 'dplyr'
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## The following objects are masked from 'package:stats':
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##
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```
## filter, lag
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## The following objects are masked from 'package:base':
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##
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## intersect, setdiff, setequal, union
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library(geosphere)
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## Warning: package 'geosphere' was built under R version 4.3.3
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#matrix distance of sampling sites of its1
dist_matrix_its1 <- distm(new.its1[,c("longitude","latitude")], fun = distHaversine)
#matrix distance of sampling sites of its2
dist_matrix_its2 <- distm(new.its2[,c("longitude","latitude")], fun = distHaversine)
```

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#Performing hierarchical clustering based on geographic distance
cluster_its1 <- hclust(as.dist(dist_matrix_its1), method = "complete")
cluster_its2 <- hclust(as.dist(dist_matrix_its2), method = "complete")

#Cutting the dendrogram into clusters using threshold of 90m
groups_its1 <- cutree(cluster_its1, h = 90) #new vector of grouped samples
groups_its2 <- cutree(cluster_its2, h = 90) #new vector of grouped samples

#Add new vector of grouped samples to datasets ####
new.its1= cbind(new.its1[,1:3], grouped_samples=groups_its1, new.its1[,4:172])
new.its2= cbind(new.its2[,1:3], grouped_samples=groups_its2, new.its2[,4:172])

#Create new permanent ID for grouped samples ####
new.ID.its1= paste0("NEF_GloFung-its1_",new.its1$grouped_samples,"_",new.its1$year)
new.ID.its2= paste0("NEF_GloFung-its2_",new.its2$grouped_samples,"_",new.its2$year)

#Add column for new ID
new.its1= cbind(new.ID= new.ID.its1, new.its1[,1:173])
new.its2= cbind(new.ID= new.ID.its2, new.its2[,1:173])

#Checking final datasets ####
length(which(new.its1[, "paper_to_keep"]=="yes",TRUE)) #1829 non-grouped samples

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## [1] 1829
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final.its1= subset(new.its1, paper_to_keep=="yes" &
  morrone_biogeoregions_Region=="Neotropical") #filtering by papers to keep
length(unique(final.its1$grouped_samples)) #466 grouped samples

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## [1] 466
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length(which(new.its2[, "paper_to_keep"]=="yes",TRUE)) #1435 non-grouped samples
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## [1] 1435
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final.its2= subset(new.its2, paper_to_keep=="yes" &
  morrone_biogeoregions_Region=="Neotropical") #filtering by papers to keep
length(unique(final.its2$grouped_samples)) #608 grouped samples

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## [1] 608
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#check grouped samples from more than 1 study
#its1
studies.its1= tapply(final.its1$paper_id, final.its1$new.ID, function(x) length(unique(x)))
n.studies.its1= studies.its1[studies.its1 > 1] #grouped samples (new.ID) including more than 1 study
#its2
studies.its2= tapply(final.its2$paper_id, final.its2$new.ID, function(x) length(unique(x)))
n.studies.its2= studies.its2[studies.its2 > 1]

#Export datasets as .csv ####

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#write.csv(new.its1, file="Data/new.its1.csv")
#write.csv(new.its2, file="Data/new.its2.csv")

#Export datasets as .csv ####
#Filter useful columns
to_use_its1= subset(new.its1, c(paper_to_keep=="yes" &
  morrone_biogeoregions_Region=="Neotropical"),select=c(new.ID,PermanentID))
to_use_its2= subset(new.its2, c(paper_to_keep=="yes" &
  morrone_biogeoregions_Region=="Neotropical"), select=c(new.ID,PermanentID))

#write.csv(to_use_its1, file="Data/short.new.its1.csv")
#write.csv(to_use_its2, file="Data/short.new.its2.csv")

#Add new datasets #####
diam= read.csv("Data/French_Guiana_soil_eDNA_metabarcoding_metadata_20.09.2024.csv", sep=",")
diam.keep= read.csv("Data/French_Guiana_soil_eDNA_metabarcoding_metadata_20.09.2024_MR.csv", sep=";")
dryf= read.csv("Data/metabarlist_fung02_for_nefineo_samples_MR.csv", sep=";")
spun= read.csv("Data/Muestras Proyecto SPUN.xlsx - Darwin Core data.csv")
mymo= read.csv("Data/Mymo_ITS_tax_forNEFINEO_seq.csv", sep=",")
long= read.csv("Data/LONGTIME_fung02_samples.csv", sep=";")

diam= cbind(diam, tokeep_NEF=diam.keep$tokeep_NEF)
diam= subset(diam, diam$tokeep_NEF=="yes")
#grouped by samples at <90m each other
dist_matrix_diam= distm(diam[,c("longitude","latitude")], fun = distHaversine)
cluster_dim= hclust(as.dist(dist_matrix_diam), method = "complete")
groups_diam <- cutree(cluster_dim, h = 90)
new.diam= cbind(plot_id=diam[,1], grouped_samples=groups_diam, diam[,2:24])
new.ID.diam= paste0("NEF_diam_",new.diam$grouped_samples,"_2025")
new.diam= cbind(new.ID=new.ID.diam, new.diam)
#write.csv(new.diam, file="Data/new.diam.csv")

#already grouped by samples at <90m each other (colname= "plot")
dryf= subset(dryf, dryf$tokeep_NEF=="yes")
new.ID.dryf= paste0("NEF_dryf_",dryf$plot,"_2025")
new.dryf= cbind(new.ID=new.ID.dryf, dryf)
#write.csv(new.dryf, file="Data/new.dryf.csv")

#already grouped by samples at <90m (colname= recordNumber)
new.ID.spun= paste0("NEF_spun_",spun$recordNumber,"_2025")
new.spun= cbind(new.ID=new.ID.spun, spun)
#write.csv(new.spun, file="Data/new.spun.csv")

#already grouped by samples at <90m (colname= SampleID)
new.ID.mymo= paste0("NEF_mymo_",mymo$SampleID,"_2025")
new.mymo= cbind(new.ID=new.ID.mymo, mymo)
#write.csv(new.mymo, file="Data/new.mymo.csv")

#already grouped by samples at <90m (colname= SampleID)
library(stringr)

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## Warning: package 'stringr' was built under R version 4.3.3
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new.ID.long= paste0("NEF_long_",str_extract(long$SampleID, "(?<=_).*"), "_2025")
new.long= cbind(new.ID=new.ID.long, long)
#write.csv(new.long, file="Data/new.long.csv")

#export to unified file showing new ID and Permanent ID of sample
a= subset(new.diam,select=c(new.ID, plot_id))
b= subset(new.dryf,select=c(new.ID, plot))
c= subset(new.spun,select=c(new.ID, recordNumber))
d= subset(new.mymo,select=c(new.ID, SampleID))
e= subset(new.long,select=c(new.ID, SampleID))
colnames(a)[2]="PermanentID"
colnames(b)[2]="PermanentID"
colnames(c)[2]="PermanentID"
colnames(d)[2]="PermanentID"
colnames(e)[2]="PermanentID"

short.new.data= rbind(a,b,c,d,e)
#write.csv(short.new.data, file="Data/short.new.data.csv")

```